



Tecnológico de Monterrey

Individual Reflection

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**Modeling of Multi-Agent Systems with
Computer Graphics**

Group 301

1. Analysis of the Developed Solution

We selected a deliberative-hybrid-reactive multi-agent model based on the functional requirements of our agricultural system. The Analyzer uses a deliberative BDI architecture because it needs to make complex decisions about plant health and maturity. The Remover combines both deliberative path planning and reactive obstacle avoidance, making it a hybrid agent suited for urgent, dynamic tasks. The Harvester follows simpler stimulus-response patterns, so a reactive architecture was sufficient and efficient. These choices were guided by variables like task complexity, environmental dynamics, communication needs, and priority levels, where infection removal is critical and harvesting is routine.

The Analyzer's complexity required it to coordinate frequently with the other agents, while the Remover's hybrid design allowed it to navigate both planned routes and sudden obstacles. Our conflict-resolution protocol, where the Remover has priority over the Harvester, emerged from the need to contain disease spread quickly. However, implementation constraints led us to simplify detection using Unity tags instead of computer vision, which affected the realism of the simulation.

We chose our graphic design for clarity and functionality. Distinct colors for plant states and simple geometric shapes made the system easy to understand and ensured smooth performance in Unity. The grid-based layout supported both agent navigation and observer comprehension, prioritizing clarity over visual realism.

The main advantages of our solution are its modular scalability, reliable coordination through AIP protocols, and effective priority-based conflict resolution. The system demonstrates MAS principles in a working prototype. However, disadvantages include simplified detection logic, dependence on the Analyzer as a central coordinator, and limited testing with scalability and dynamic environmental changes.

To address these limitations, we could integrate actual image processing with Unity's ML-Agents, distribute perception capabilities across agents, enhance obstacle avoidance

algorithms, and support multiple Analyzers to reduce single-point failure risk, and rigorous scalability under heavier loads.

2. Reflection on Learning Process

Looking back at my initial expectations, I see a significant evolution between what I hoped to learn and what I actually experienced. I fully expected to design and build a multi-agent simulation in Unity and Blender, and I am proud to say we delivered that, a functional greenhouse environment with three coordinated agents. I also anticipated applying algorithms to make the system intelligent, and while we successfully implemented agent coordination and decision-making logic, we could not integrate the advanced computer vision or complex AI models I initially envisioned. This was due to significant time constraints and a critical lack of resources; specifically, I was unable to run the Unity project on my personal computer due to insufficient hardware capacity, which limited my ability to test and implement more computationally intensive features.

Despite this hardware limitation, I successfully contributed to the logical design, protocol development, and documentation. This adaptation taught me a crucial lesson in practical project development: the core intelligence of a system often lies in its architecture and coordination, not just in advanced algorithms. My expectation to deliver a useful project for the client was met in spirit, we built a working prototype that demonstrates a viable automation concept, even if it's not yet a production-ready tool.

This entire journey strengthened my foundational skills in multi-agent system design and team collaboration. I learned that effective problem-solving often means prioritizing a working, integrated system over technically ambitious but unimplementable features. While I didn't get to explore all the advanced topics I hoped for, I gained deep, practical experience in turning a theoretical agent-based design into a cohesive, running simulation, a different but equally valuable form of learning.